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THE HIT-AND-RUN IN FATAL PEDESTRIAN ACCIDENTS: VICTIMS, CIRCUMSTANCES AND DRIVERS

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Abstract—Hit-and-run pedestrian fatalities are classified as to victim characteristics, accident circumstances and driver characteristics. Over 18,000 pedestrian fatalities are reported in the Fatal Accident Recording System for 1989– 1991. Twenty percent of the drivers involved left the accident scene. Children and senior citizens are the least likely to be left. Drivers more commonly run in urban areas, outside the South, on weekends and at night. Comparing motorists who left the scene but were later identified with the drivers who remained, drivers aged 66 and older are half as likely as younger drivers to leave the accident scene and male drivers are 60% more likely than female drivers to run. Drivers with previous driving while intoxicated arrests are twice as likely to run as those with no such arrests. Forty-nine percent of hit-and-run motorists involved in pedestrian fatalities are ultimately identified by the police. Other factors constant, drivers are more likely to be identified if the victim was female or a child. Drivers are identified 2.5 times as often when the incident occurred in daylight. Drivers appear to run more often when they are at fault and will be severely punished (e.g. when they are intoxicated) and when it is likely that they can escape detection (e.g. at night).

Keywords—Pedestrian fatalities, Hit-and-run, Drivers

INTRODUCTION

Leaving the scene of an accident is a crime. The hit-and-run is an interesting crime because it is typically spontaneous and not committed by career criminals. It is also a crime that has received virtually no research attention.

This paper examines the hit-and-run pedestrian fatality. Analyses of pedestrian injuries sometimes mention the proportion caused by hit-and-run drivers (Haddon et al. 1961; Baker et al. 1974; Knoblauch 1977; Rivara and Barber 1985), but none has explored the factors involved in these incidents. The present study provides an epidemiology of the hit-and-run pedestrian fatality in terms of victim characteristics, accident circumstances and driver characteristics.

METHODS

Data are based on every pedestrian fatality in the United States (excluding Alaska and Hawaii) during 1989, 1990 and 1991, taken from the Fatal Accident Reporting System (FARS) of the National Highway Traffic Safety Administration. The police are required to submit information on all fatal accidents.

The FARS data are organized into three separate files: person, vehicle and accident. Code numbers link data in the three files, making it possible to assemble a composite picture of the participants and vehicles involved in each incident. For each pedestrian fatality, identification of the accident was used to obtain the time of the event and whether it was a hit-and-run. Identification of the striking vehicle was used to retrieve the information on that car and its driver.

The dependent variables are whether the collision was a hit-and-run, as coded by the police at the scene of the accident, and whether the hit-and-run driver was subsequently identified by the police. Data on the driver's presence at the scene are considered to be very accurate. A driver is classified as identified if the FARS data file includes the driver's age.

We focus on three groups of independent variables: victim characteristics, circumstances of the collision, and driver characteristics. Information on the victim's age and sex is nearly complete, as are many variables on the circumstances of the crash: time of day, day of week, month, region of the country,

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population density, prevailing speed limit, and light conditions. Time of day and day of the week are used to separate accidents into weekend, defined as 6 p.m. on Friday until 6 p.m. on Sunday, and weekday events.

Information on hit-and-run drivers and their driving histories is available only for drivers who have been identified by the police. FARS data on the driver include gender, age, age of the car (divided into new and old– whether the model year is within 5 years of the time of the incident) and driving record (validity of the driver's license and the presence of previous driving-while-intoxicated (DWI) convictions). A final variable is the driver's blood alcohol content (BAC) following the collision (divided into three groups: positive BAC, zero BAC and no test given). BAC tests are not usually administered, they are not performed at random, and they may be given too late to detect the driver's level of intoxication at the time of the crash.

Simple cross tabulations are used to discover how accident circumstances and the characteristics of victim and driver affect the proportion of hit-and-runs and how circumstances and victim characteristics affect the proportion of hit-and-run drivers who are subsequently identified. Significance levels are determined by chi-squared tests. We use multiple logistic regression to examine the effect of each variable controlling for other factors. When examining the factors affecting the probability of hitting-and-running, we present two regressions: one using all pedestrian fatalities and one using only observations for which driver characteristics are available.

RESULTS

In the United States, over 18,000 pedestrians were killed from 1989 to 1991. Twenty percent of the drivers involved left the accident scene.

Victims

The rate of hit-and-run incidents varies significantly with the victim's age (Table 1). Children and the elderly are the least likely to be left–only 10% of motorists run after killing someone younger than 6 or older than 80. Holding circumstances and driver variables constant, motorists are half as likely to run after killing a person over 65 (odds ratio (OR) = 0.54, 95% confidence interval (CI) = [0.45, 0.63]) (Table 2) and 80% as likely to run when the victim is under age 11 (OR = 0.81, 95% CI = [0.66, 0.99]) (not shown). The pedestrian's sex does not affect the probability of being the victim of a hit-and-run when the circumstances of the collision are controlled for (Table 2).

Circumstances

Drivers more commonly leave the accident scene in urban areas, during the summer months, on weekends and on roads where the speed limit is lower than 55 mph. Hitting-and-running is less frequent in the South and in daylight (Table 1). All these results hold in the multivariate analysis (Table 2). For example, when the circumstances of the collision and the characteristics of victim and motorist are all held constant, leaving the accident scene is still 50% more likely on weekends (OR = 1.50, 95% CI = [1.34, 1.67]) and 40% less likely in daylight (OR = 0.58, 95% CI = [0.49, 0.69]) (Table 2).

Drivers

Analysis of the influence of driver characteristics on the propensity to hit-and-run is limited to comparing drivers who remained at the scene to hit-and-run drivers who were ultimately identified by the police. Based on this sample, the driver's age and sex have a significant impact on the likelihood of hitting-and-running. Holding other factors constant, drivers aged 66 and older are half as likely as younger drivers to leave the accident scene (OR = 0.48, 95% CI = [0.34, 0.67]) (Table 2). Male drivers are 60% more likely than female drivers to run (OR = 1.63, 95% CI = [1.41, 1.89]).

Drivers of new cars are less likely to run compared to drivers of older vehicles (OR = 0.54, 95% CI = [0.49, 0.61]). An invalid license, a previous DWI conviction and a positive BAC each more than double the probability that a pedestrian fatality is a hit-and-run.

Forty-nine percent of hit-and-run motorists involved in pedestrian fatalities are ultimately identified by the police (Table 3). Other factors constant, drivers are more likely to be identified if the victim is a child (OR = 2.12, 95% CI = [1.60, 2.81]) and less likely to be identified if the victim is male (OR = 0.75, 95% CI = [0.63, 0.88])(Table 4). When hit-and-runs occur outside the South, the driver is identified more often. Although it appears that drivers are more commonly identified when the speed limit is under 55 mph (Table 3), this effect vanishes when other factors are held constant (Table 4). The most important factor associated with the driver's identity being discovered is the prevailing light conditions. Drivers are identified 2.5 times as often when the incident occurred in daylight (OR = 2.55, 95% CI = [1.98, 3.28]) (Table 4).

DISCUSSION

Of all pedestrian fatalities in the United States from 1989 to 1991, 20% of drivers left the scene.

Table 1. Correlates of hit-and-run in pedestrian fatalities

	N	% hit-and-run		N	% hit-and-run
Overall	18,821	20	*Light conditions*		
Characteristics of victims			Daylight	6223	9**
Victim's age			Dark	5975	22
0–5	985	10**	Other[4]	6566	27
6–10	1014	11	**Characteristics of drivers[5]**		
11–15	694	18	*Driver's age*		
16–20	1107	25	16–20	2370	11**
21–25	1142	29	21–25	2761	13
26–30	1577	27	26–30	2589	12
31–35	1679	26	31–35	2087	10
36–45	2534	25	36–45	2776	8
46–55	1719	21	46–55	1639	5
56–60	868	17	56–60	567	6
61–65	874	16	61–65	497	6
66–70	850	12	66–70	371	3
71–75	897	13	71 and older	695	5
76–80	988	11	Unknown	2667	82
81 and older	1343	9	*Driver's sex*		
Victim's sex			Male	12,767	12**
Male	13,062	21**	Female	3936	6
Female	5751	17	Unknown	2118	88
Circumstances			*Car age*		
Region			5 years old or less	7905	8**
South[1]	6701	18**	More than 5 years old	8885	13
Other	12,120	21	Unknown	2031	96
Population density			*Driver's license status*		
Urban	13,261	21**	Valid	14,600	7**
Rural	5510	16	Invalid	1638	29
Season			Unknown	2583	84
Summer[2]	4563	21*	*Previous DWI conviction (past 3 years)*		
Other	14,258	19	No	15,618	9**
Time of week			Yes	510	29
Weekend[3]	6818	25**	Unknown	2693	82
Weekday	11,904	16	*Driver alcohol involvement*		
Speed limit			Zero BAC	2369	4**
Less than 55 mph	12,777	20*	Positive BAC	1617	20
55 or more mph	5335	18	Other[6]	14,835	22

[1]South consists of AL, AR, FL, GA, KY, LA, MS, NC, SC, TN, TX, VA and WV.
[2]Summer consists of June, July and August.
[3]Weekend is defined as 6 p.m. Friday until 6 p.m. Sunday.
[4]Other consists of dawn, dusk, and dark but lighted.
[5]Significance tests are performed excluding "Unknown" category.
[6]Other includes test not given or refused as well as unknown. *$p<0.01$; **$p<0.001$.

Almost half of the hit-and-run motorists were ultimately identified.

A few studies of pedestrian injuries report the proportion of hit-and-runs. For example, 4 out of 50 adult pedestrian fatalities (8%) in Manhattan in 1959 were caused by hit-and-run drivers (Haddon et al. 1961). Of the 182 pedestrian fatalities occurring from 1968 to 1971 in Baltimore, 13% were caused by hit-and-run drivers, of whom 88% were eventually identified (Baker et al. 1974). In an analysis of 1531 rural and suburban pedestrian accidents from 1974, 9% were hit-and-runs (Knoblauch 1977). A study of all pedestrian injuries to children in Memphis in 1982 found that 15% of drivers left the scene (Rivara and Barber 1985). The circumstances and characteristics of participants in these incidents have never been examined in depth.

Victims

Drivers are much less likely to leave the scene when the victim is either quite young–under 11 years old–or of advanced age–over 65 years old. It may be that drivers feel more guilt or more responsibility when they injure people in these age groups. In addition, most drivers who kill very young pedestrians may not be strangers (Mueller et al. 1987).

Younger and older pedestrians may be less likely to be killed at times and in places conducive to leaving the scene. Children and the elderly are most likely to be victims during daylight hours (not shown). But even holding the time of day and other factors constant, drivers are less likely to leave victims in these age categories.

Drivers may also be less likely to be legally culpable when they fatally injure victims who are very

Table 2. Effects of characteristics of victims, circumstances and drivers on probability of hit-and-run after pedestrian fatality: logistic regression results

	I (N = 17,713)		II (N = 15,844)	
	Odds ratio	95% confidence interval	Odds ratio	95% confidence interval
Victim characteristics				
Victim 0–15	0.72	(0.63, 0.83)	0.92	(0.78, 1.09)
Victim 16–25	1.23	(1.10, 1.36)	1.32	(1.14, 1.52)
Victim ≥66	0.52	(0.46, 0.58)	0.54	(0.45, 0.63)
Male victim	1.06	(0.97, 1.15)	0.92	(0.82, 1.04)
Circumstances				
South	0.77	(0.70, 0.84)	0.63	(0.56, 0.72)
Urban	1.37	(1.24, 1.52)	1.31	(1.13, 1.51)
Summer	1.20	(1.09, 1.31)	1.15	(1.02, 1.29)
Weekend	1.51	(1.39, 1.63)	1.50	(1.34, 1.67)
Speed limit ≥55 mph	0.79	(0.71, 0.87)	0.78	(0.68, 0.90)
Daylight	0.43	(0.38, 0.49)	0.58	(0.49, 0.69)
Other light conditions	1.17	(1.06, 1.29)	1.15	(1.00, 1.32)
Driver characteristics				
Driver 16–25			1.07	(0.91, 1.25)
Driver 26–30			0.99	(0.88, 1.12)
Driver ≥66			0.48	(0.34, 0.67)
Male driver			1.63	(1.41, 1.89)
Car ≤5 years old			0.54	(0.49, 0.61)
Invalid license			2.81	(2.44, 3.23)
Previous DWI conviction(s)			2.05	(1.63, 2.57)
Positive BAC			3.20	(2.49, 4.11)
BAC unknown			2.74	(2.20, 3.41)

young. Children's road behavior is often deficient (Malek et al. 1990). Since children are more likely to have been at fault in the collision, drivers may be more confident that they will not face prosecution.

Drivers who run after hitting a young pedestrian are also most likely to be identified. Perhaps the drivers may experience more remorse and give themselves up voluntarily or police may more vigorously pursue the hit-and-run drivers in accidents involving children. Police may be better able to identify such individuals because when a child is struck there are often witnesses in the vicinity; child pedestrians are usually injured while playing near their homes (Malek et al. 1990). Overall, only 3% of drivers in child (10 years old and younger) pedestrian fatalities are unidentified motorists.

When a male pedestrian is killed, a higher percentage of drivers flee the scene. However, the association of hit-and-run with victim gender is due entirely to the circumstances of the accident. When other factors are taken into account, female pedestrian fatalities are equally likely to be hit-and-runs, but the motorists responsible are more likely to be subsequently identified by police.

Circumstances

Knoblauch (1977) found that drivers were equally likely to leave the scene in urban and rural pedestrian accidents. In our analysis, drivers run more often in urban areas. More than two decades of research has established that a stranger's need or request for assistance is more likely to get a response in rural areas than in urban areas (Steblay 1987; Amato 1993). It is particularly interesting that this pattern was observed in four studies in which the stranger was an injured pedestrian (Steblay 1987).

Motorists in the South are less likely to hit-and-run, perhaps because of the supposed Southern emphasis on manners and the Southern tradition of helping others (Wilson 1989). Southerners consume less alcohol per capita than residents of other regions and are more likely to abstain from drinking (Williams et al. 1992). If drivers do leave the scene in the Southern states, however, they have only two-thirds the chance of becoming known to the police compared to hit-and-run drivers in other parts of the country, even when other circumstances of the collision are taken into account.

Some studies have suggested that pedestrian injuries vary seasonally, occurring less frequently in the summer (Read et al. 1963; Vestrup and Reid 1989). Another study (Knoblauch 1977), and our data, show no seasonal variation in the rate of pedestrian accidents. However, we found that drivers more commonly leave the scene in the summer, both in the simple cross-section and when controlling for related factors such as whether the collision occurred in daylight.

Light conditions are a crucial factor both in whether the driver leaves the scene and whether the police ever identify the driver. Hit-and-runs are least likely to occur in daylight and such hit-and-run motorists are most likely to be discovered. Consequently, only 3% of daylight pedestrian fatalities are caused by drivers who are never identified. Drivers may decide to remain at the scene when the fatality occurs in daylight because they realize that the chance of escaping detection is so low.

Drivers are more likely to leave the scene on weekends. The association of hit-and-runs with nights, weekends and summers may well be an effect of alcohol (Solnick and Hemenway 1994).

Drivers

When exploring the influence of driver characteristics on the probability of committing a hit-and-run, we are limited to comparing drivers who stayed at the scene of the accident with those who left and were later identified. Those identified may differ from hit-and-run drivers who escape detection–indeed, we have discovered that certain accident circumstances

Table 3. Correlates of hit-and-run drivers identified

	N hit-and-run	% H-and-R drivers identified	N pedestrian fatalities	% Ped. fatals unidentified hit-and-run
Overall	3686	49	18,821	10
Victim characteristics				
Victim's age				
0–5	96	76**	985	2**
6–10	111	70	1014	3
11–15	125	70	694	5
16–20	279	50	1107	12
21–25	423	50	1452	15
26–30	427	42	1577	16
31–35	438	46	1679	14
36–45	622	48	2534	13
46–55	366	45	1719	12
56–60	147	51	868	8
61–65	137	42	874	9
66–70	106	49	850	6
71–75	113	54	897	6
76–80	112	50	988	6
≥ 81	117	49	1343	4
Victim's sex				
Male	2705	47**	13,062	11**
Female	979	57	5751	7
Circumstances				
Region				
South	1199	41**	6701	11
Other	2487	53	12,120	10
Population density				
Urban	2805	50	13,261	10**
Rural	873	47	5510	8
Season				
Summer	955	51	4563	10
Other	2731	49	14,258	10
Time of week				
Weekend	1710	50	6818	13**
Weekday	1885	51	11,904	8
Speed limit				
< 55 mph	2529	52**	12,777	10
≥ 55 mph	942	46	5335	10
Light conditions				
Daylight	568	70**	6223	3**
Dark	1307	43	5975	12
Other	1778	47	6566	14

**$p < 0.001$.

Table 4. Effects of victim characteristics and circumstances on probability of hit-and-run driver being identified: logistic regression results ($N = 3306$)

	Odds ratio	95% confidence interval
Victim characteristics		
Victim 0–15	2.12	(1.60, 2.81)
Victim 16–25	1.17	(0.97, 1.40)
Victim ≥ 66	0.89	(0.71, 1.12)
Male victim	0.75	(0.63, 0.88)
Circumstances		
South	0.66	(0.57, 0.77)
Urban	0.91	(0.75, 1.12)
Summer	1.07	(0.91, 1.26)
Weekend	1.13	(0.98, 1.31)
Speed limit ≥ 55 mph	0.95	(0.79, 1.15)
Daylight	2.55	(1.98, 3.28)
Other light conditions	1.08	(0.90, 1.29)

and victim characteristics influence the probability that a driver is traced. Nevertheless, since any sample of hit-and-runs will include accidents caused by unknown drivers, it is still useful to compare driver characteristics in the two groups for which data are available.

Drivers older than 65 years are only half as likely to leave the scene as younger drivers. Young drivers evaluate the potential negative consequences of reckless driving behavior less negatively and the potential positive consequences (e.g. arriving at the destination more quickly) more positively than older drivers (Parker et al. 1992). They feel less pressure from peers to avoid committing driving violations (Parker et al. 1992). Youth are also more likely to drive after drinking (Donovan et al. 1983, 1985; Macdonald and Pederson 1990; Wilson 1992).

While the victim's gender does not affect the probability of a hit-and-run, the driver's gender does. Many more male drivers kill pedestrians and men are more likely to leave the scene. This result is not surprising, as men are more likely to commit other driving violations, such as speeding (Galin 1981; Hemenway and Solnick 1993; but see Wasielewski 1984) and driving after drinking (Donovan et al. 1983; Richman 1985; Anda et al. 1987; Bradstock et al. 1987; Macdonald and Pederson 1990; McMillen et al. 1991; Wilson 1992; Hemenway and Solnick 1993).

Having an invalid license or previous DWI conviction more than doubles the chance that a driver will run from a pedestrian fatality. In a study of pedestrian trauma, 88% of the drivers for whom records could be obtained had a previous accident (Vestrup and Reid 1989). Not only are these drivers more likely to strike a pedestrian, but our results suggest they are more likely to drive away afterwards.

In our previous analysis of pedestrian fatalities, we argue that alcohol involvement is an important factor in the probability of leaving the scene of a crash (Solnick and Hemenway 1994). The time of the accident is available for all pedestrian fatalities. The increased proportion of hit-and-runs on the weekend and at night, periods when drivers are more likely to be drinking (Heeren et al. 1985; Fatal Accident Reporting System 1991), indicates that alcohol is associated with a greater propensity to hit-and-run.

Alcohol also appears as an important influence on leaving the accident scene when the analysis is restricted to drivers for whom information is available. Identified hit-and-run drivers are more likely to be young and male, to live outside the South and to have invalid licenses or previous DWI convictions. For drivers tested, those who left the scene were three times as likely to have positive BAC.

Drivers who have been drinking may be more likely to hit-and-run for a variety of reasons. Such drivers are likely to be responsible for the crash (Baker et al. 1974; Mounce and Pendleton 1992). In addition, alcohol has been found to compromise moral judgment (Steele and Southwick 1985; Denton and Krebs 1990) and increase aggressiveness (Taylor et al. 1976; Pernanen 1991), making drivers more willing to drive away and avoid punishment. Drunk drivers may rightly fear that the law will treat them harshly because they were intoxicated (Theroux 1985). Alternatively, drivers who break the law by driving drunk may also be the type who are more inclined to break the law by leaving the scene of an accident.

The analysis has certain limitations. Drivers hit-and-run in many situations other than pedestrian fatalities. However, limiting our attention to one type of collision allows us to control for factors that vary among different kinds of accidents. The accidents we focus on are quite serious and deserving of study. Pedestrian injuries have received less attention than injuries to motor vehicle occupants (Vestrup and Reid 1989; Malek et al. 1990).

When analyzing how driver characteristics influence the probability of hitting-and-running, we are restricted to drivers whose identities were later discovered by the police. While other research assumes that a random subset of drivers were traced (Theroux 1985), we find that hit-and-runs do not have an equal chance of being solved: when a pedestrian fatality occurs in daylight or outside of the South or when the victim is female or young, the hit-and-run driver is more likely to be identified.

While rarely premeditated, hit-and-runs are not random. Accident circumstances and characteristics of victim and driver all affect the likelihood of running after a pedestrian fatality. Certain types of victim–children and senior citizens–are least likely to be left at the accident scene. And certain types of drivers—those who are older, female, living in the South or with clean driving records—are less likely to hit-and-run.

Running appears to be related to whether the driver feels he can escape detection. For example, drivers are more likely to run at night. Drivers are also more likely to run in circumstances suggesting they were at fault. Thus hit-and-runs are common when motorists are likely to be intoxicated (e.g. on weekends, at night and for drivers with previous DWI convictions). Drunk drivers are also likely to face severe penalties.

Both the difficulty of escape and the possible absence of legal culpability may help explain why so few drivers run when the victim is a child. Since there may be witnesses nearby or the driver may be known to the family, drivers may not feel they can get away. And if the child contributed to the accident by poor road behavior, the driver may feel there will be few negative legal consequences of remaining at the scene.

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REFERENCES

Amato, P. R. Urban-rural differences in helping friends and family members. Soc. Psychol. Q. 56:249–262; 1993.

Anda, R. F.; Remington, P. L.; Dodson D. L.; DeGuire, P.J.; Forman, M. R.; Gunn, R. A. Patterns of self-reported drinking and driving in Michigan. Am. J. Prev. Med. 3:271–275; 1987.

Baker, S. P.; Robertson, L. S.; O'Neill, B. Fatal pedestrian collisions: Driver negligence. Am. J. Public Health 64:318–325; 1974.

Bradstock, M. K.; Marks, J. S.; Forman M. R.; Gentry, E. M.; Hogelin, G. C.; Binkin, N. J.; Trowbridge, F. L. Drinking-driving and health lifestyle in the United States: Behavioral risk factors surveys. J. Stud. Alcohol 48:147–158; 1987.

Denton, K.; Krebs, D. From the scene to the crime: The effect of alcohol and social context on moral judgement. J. Pers. Soc. Psychol. 59:242–248; 1990.

Donovan, D. M.; Marlatt, G. A.; Salzberg, P. M. Drinking behavior, personality factors and high-risk driving: A review and theoretical formulation. J. Stud Alcohol 44:395–428; 1983.

Donovan, D. M.; Queisser, H. R.; Salzberg, P. M.; Umlauf, R. L. Intoxicated and bad drivers: Subgroups within the same population of high risk men drivers. J. Stud. Alcohol 46:375–397; 1985.

Fatal Accident Reporting System (FARS). DOT Publication No. HS-807-693. Washington: National Highway Traffic Safety Administration, U.S. Department of Transportation; March 1991.

Galin, D. Speed on two-lane rural roads-a multiple regression analysis. Traffic Eng Control 22:453–460; 1981.

Haddon, W. Jr.; Valien, P.; McCarroll, J. R.; Umberger, C. J. A controlled investigation of the characteristics of adult pedestrians fatally injured by motor vehicles in Manhattan. J. Chron. Dis. 14:655–678; 1961.

Heeren, T.; Smith, R. A.; Morelock, S.; Hingson, R. W. Surrogate measures of alcohol involvement in fatal crashes: Are conventional indicators adequate? J. Saf. Res. 16:127–134; 1985.

Hemenway, D.; Solnick, S. J. Fuzzy dice, dream cars, and indecent gestures: Correlates of driver behavior? Accid. Anal. & Prev. 25:161–170; 1993.

Knoblauch, R. L. Causative factors and countermeasures for rural and suburban pedestrian accidents: Accident data collection and analysis, DOT HS-802 266, Washington, DC: Department of Transportation, National Highway Traffic Safety Administration; 1977.

Macdonald, S.; Pederson, L. L. The characteristics of alcoholics in treatment arrested for driving while impaired. Br. J. Addict. 85:97–105; 1990.

Malek, M.; Guyer, B.; Lescohier, I. The epidemiology and prevention of child pedestrian injury. Accid. Anal. & Prev. 22:301–313; 1990.

McMillen, D. L.; Pang, M. G.; Wells-Parker, E.; Anderson, B. J. Behavior and personality traits among DUI arrestees, nonarrested impaired drivers and nonimpaired drivers. Int. J. Addict. 26:227–235; 1991.

Mounce, N. H.; Pendleton, O. J. The relationship between blood alcohol concentration and crash responsibility for fatally injured drivers. Accid. Anal. & Prev. 24:201–210; 1992.

Mueller, B. A.; Rivara, F. P.; Bergman, A. B. Factors associated with pedestrian-vehicle collision injuries and fatalities. West. J. Med. 146:243–245; 1987.

Parker, D.; Manstead, A. S. R.; Stradling, S. G.; Reason, J. T. Determinants of intention to commit driving violations. Accid. Anal. & Prev. 24:117–131; 1992.

Pernanen, K. Alcohol in human violence. New York, NY: Guilford Press; 1991.

Read, J. H.; Bradley, E. J.; Morison, J. D.; Lewall, D.; Clarke, D. A. The epidemiology and prevention of traffic accidents involving child pedestrians. Can. Med. Assoc. J. 89:687–701; 1963.

Richman, A. Human risk factors in alcohol-related crashes. J. Stud. Alcohol 10 (Suppl):21–31; 1985.

Rivara, F. P.; Barber, M. Demographic analysis of childhood pedestrian injuries. Pediatrics 76:375–381; 1985.

Solnick, S. J.; Hemenway, D. Hit the bottle and run: The role of alcohol in hit-and-run pedestrian fatalities. J. Stud. Alcohol 55:679–684; 1994.

Steblay, N. M. Helping behavior in rural and urban environments: A meta-analysis. Psychol. Bull. 102:346-356; 1987.

Steele, C. M.; Southwick, L. Alcohol and social behavior I: The psychology of drunken excess. J. Pers. Soc Psychol. 48:18–24; 1985.

Taylor, S. P.; Gammon, C. B.; Capasso, D. R. Aggression as a function of the interaction of alcohol and threat. J. Pers. Soc. Psychol. 34:938–939; 1976.

Theroux, R. P. The effects of Ohio's 1983 driving while intoxicated laws. Unpublished Ph.D. dissertation, Purdue University; 1985.

Vestrup, J. A.; Reid, J. D. S. A profile of urban adult pedestrian trauma. J. Trauma 29:741–745; 1989.

Wasielewski, P. Speed as a measure of driver risk: Observed speed versus driver and vehicle characteristics. Accid. Anal. & Prev. 16:89–103; 1984.

Williams, G. D.; Stinson, F. S.; Clem, D.; Noble, J. Apparent per capita alcohol consumption: National, state, and regional trends, 1977–1990. National Institute on Alcohol Abuse and Alcoholism, Surveillance Report #23; 1992.

Wilson, C.R.; "Manners", In: Wilson, C. R.; Ferris, W. Eds. Encyclopedia of Southern Culture. Chapel Hill, NC: University of North Carolina Press; 1989: pp. 634–637.

Wilson, R. J. Convicted impaired drivers and high risk drivers: How similar are they? J. Stud. Alcohol 53:335–344; 1992.